

Group Theory

2. Group Multiplication Table (The Cayley Table)

All the information about any discrete group is provided by its multiplication table (also called a Cayley table). Let $G = \{e \equiv a_0, a_1, a_2, \dots, a_{n-1}\}$ be a (finite) group of order n . We construct $n \times n$ table which each row and each column is labelled by a group element in the above order.

A typical multiplication table

	e	a_1	a_2	...	a_{n-1}
e	e	a_1	a_2	...	a_{n-1}
a_1	a_1				
a_2	a_2				
..	..				
..	..				
a_{n-1}	a_{n-1}				